

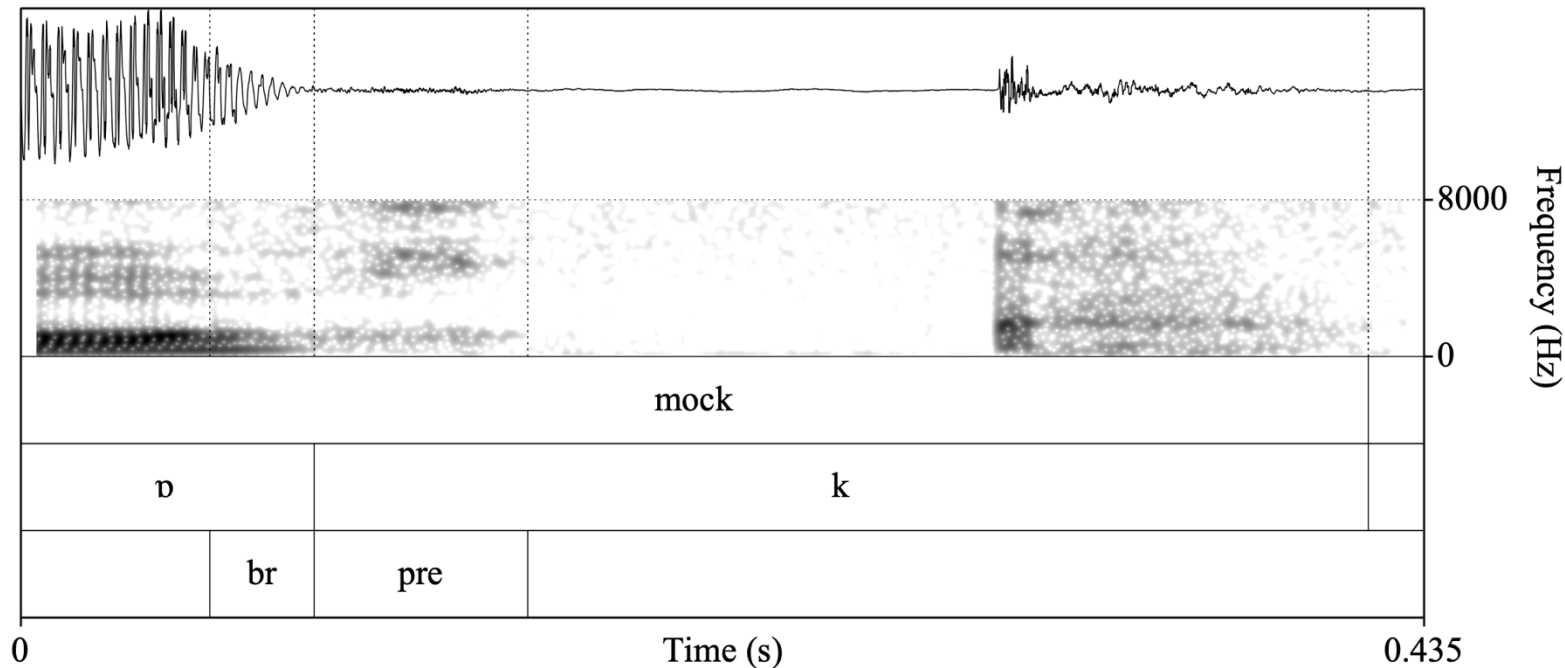
Toward modelling indirect effects in longitudinal phonetics research: The case of Queen Elizabeth II's age, voice quality, and pre-aspiration

Thomas Kettig¹, Míša Hejrná², Noah Henry³

¹York University, ²Aarhus Universitet, ³Universität Innsbruck

Pre-aspiration

- **Glottal friction** that may occur in sequences of sonorants and phonetically voiceless obstruents
 - Welsh English *mock* [mp^hk^h]

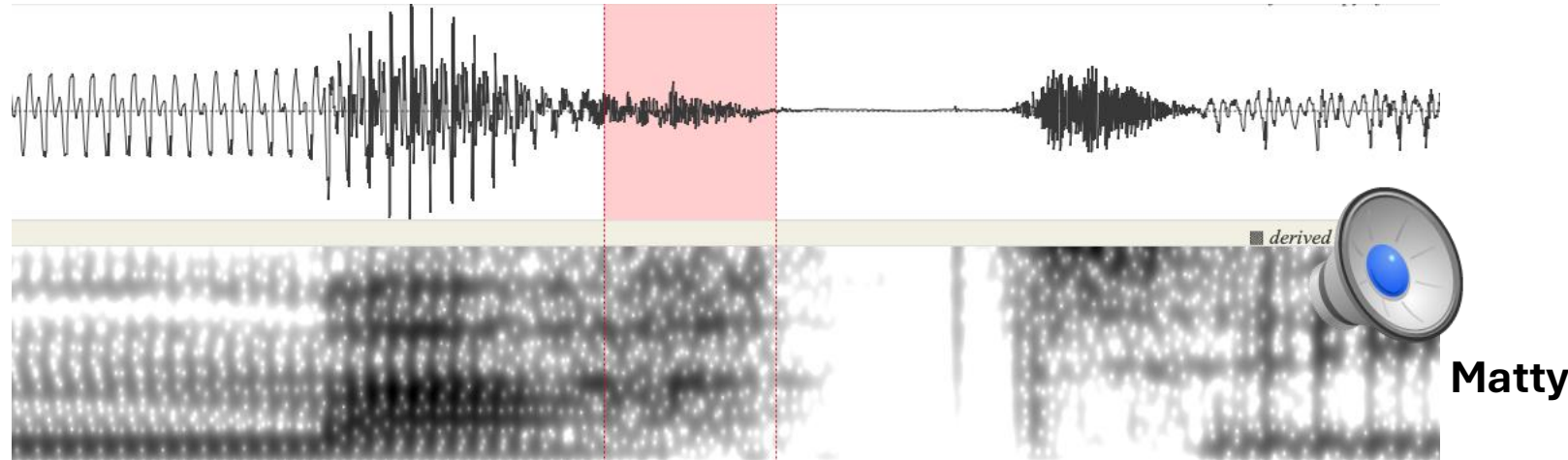
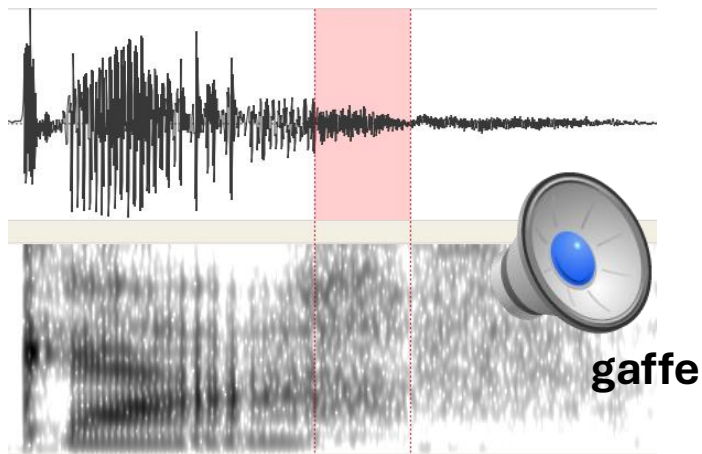


Pre-aspiration

- Favoured:
 - by low vowels (vs high)
 - by phonologically short vowels (vs long)
 - by posterior places of articulation (vs anterior: /k, t, p/)
 - by prosodically prominent positions (lexical and other types of prominence)
 - (e.g. Hejrná 2025)
- Noted in an **increasing number** of English accents in the UK, North America, and Aus/NZ

Pre-aspiration

- Kettig (2015) found it all over in age 18–24 SSBE speakers when attempting to measure the duration of various /æ/ words



- Is it actually on the rise? Or are we just looking more closely at speech and finding it where it was rarely observed before?

Our case study: QEII

- If she pre-aspirated, then it's not innovative in SSBE
- Maybe there's age grading: are younger people more pre-aspirating?
- We have lots of data from her: Christmas speeches provide a single-style longitudinal dataset from 1952 onwards
- Can investigate both age- and phonological environment-related variation in pre-aspiration



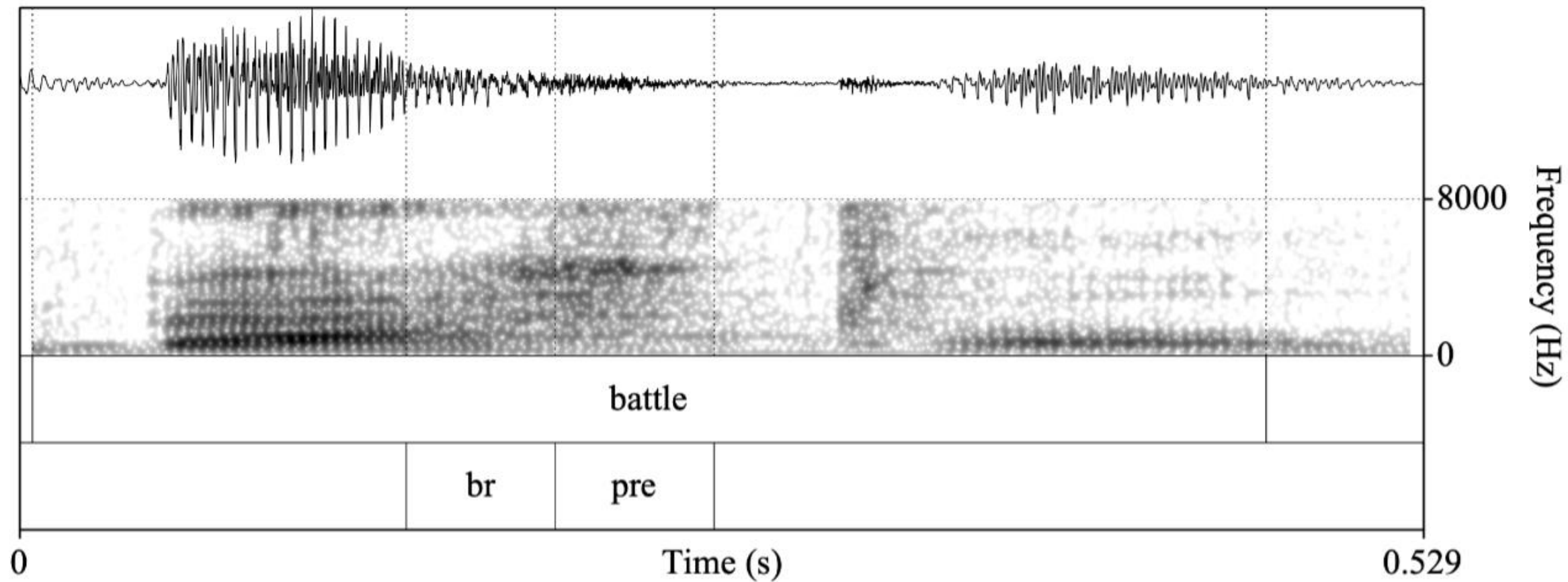
pre-aspirated “battle”

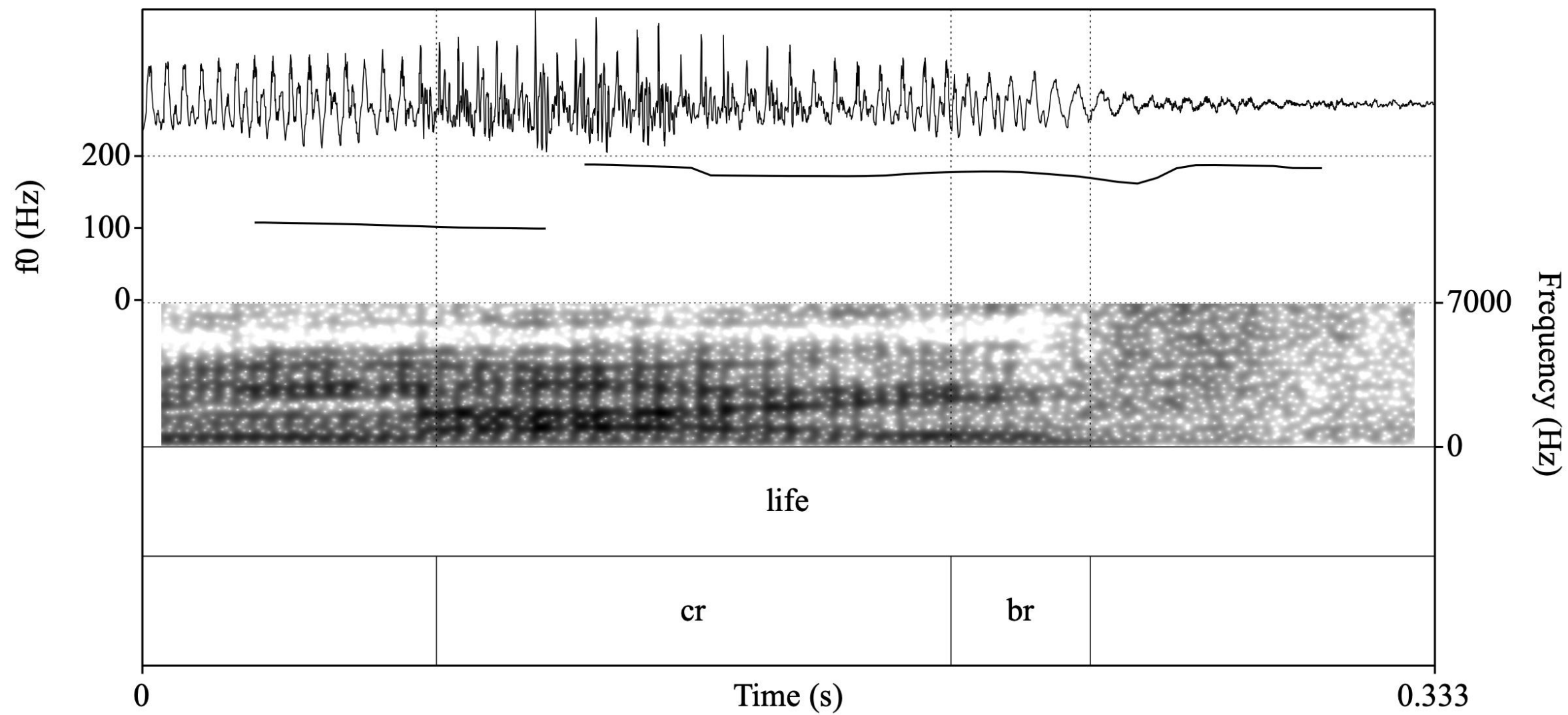
Relationship to voice quality

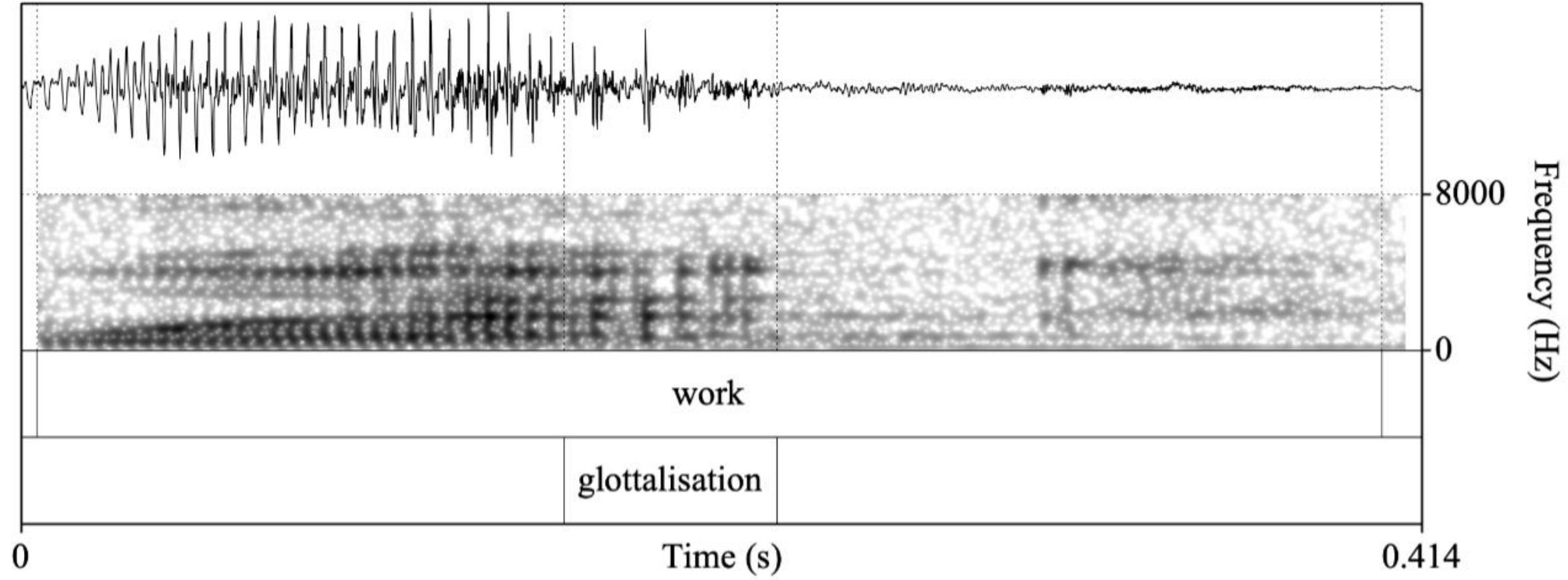
- Pre-aspiration (almost) always accompanied by local breathy voice, only rarely accompanied by glottalization (creak)
- Breathiness and glottalization can be subject to agentive/social variation, but also physiological aging processes
 - Mixed findings in the lit regarding increasing creakiness with age (e.g. Gittelson et al. 2021, Grama et al. 2023)
 - But no (?) longitudinal studies on local phonatory variation – all research seems to be apparent-time
- So how might breathiness and glottalization relate to pre-aspiration? Maybe age affects voice quality, which in turn feeds or bleeds pre-aspiration!

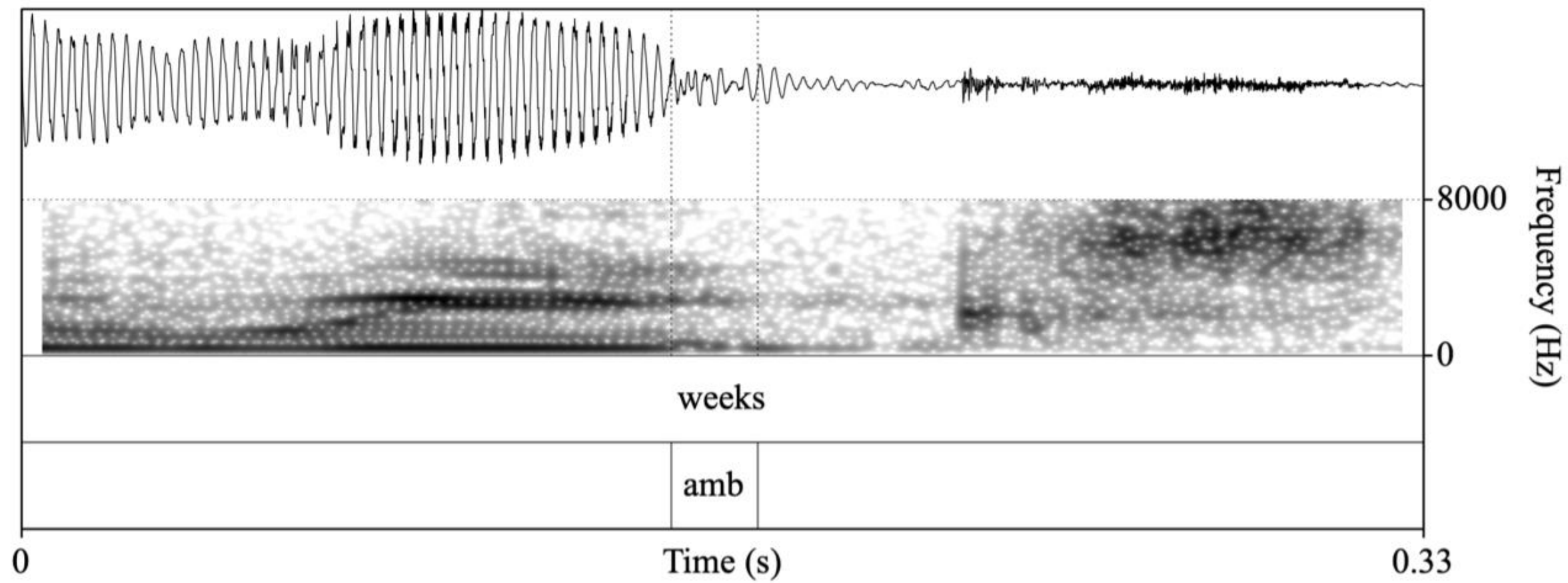
Coding acoustic data

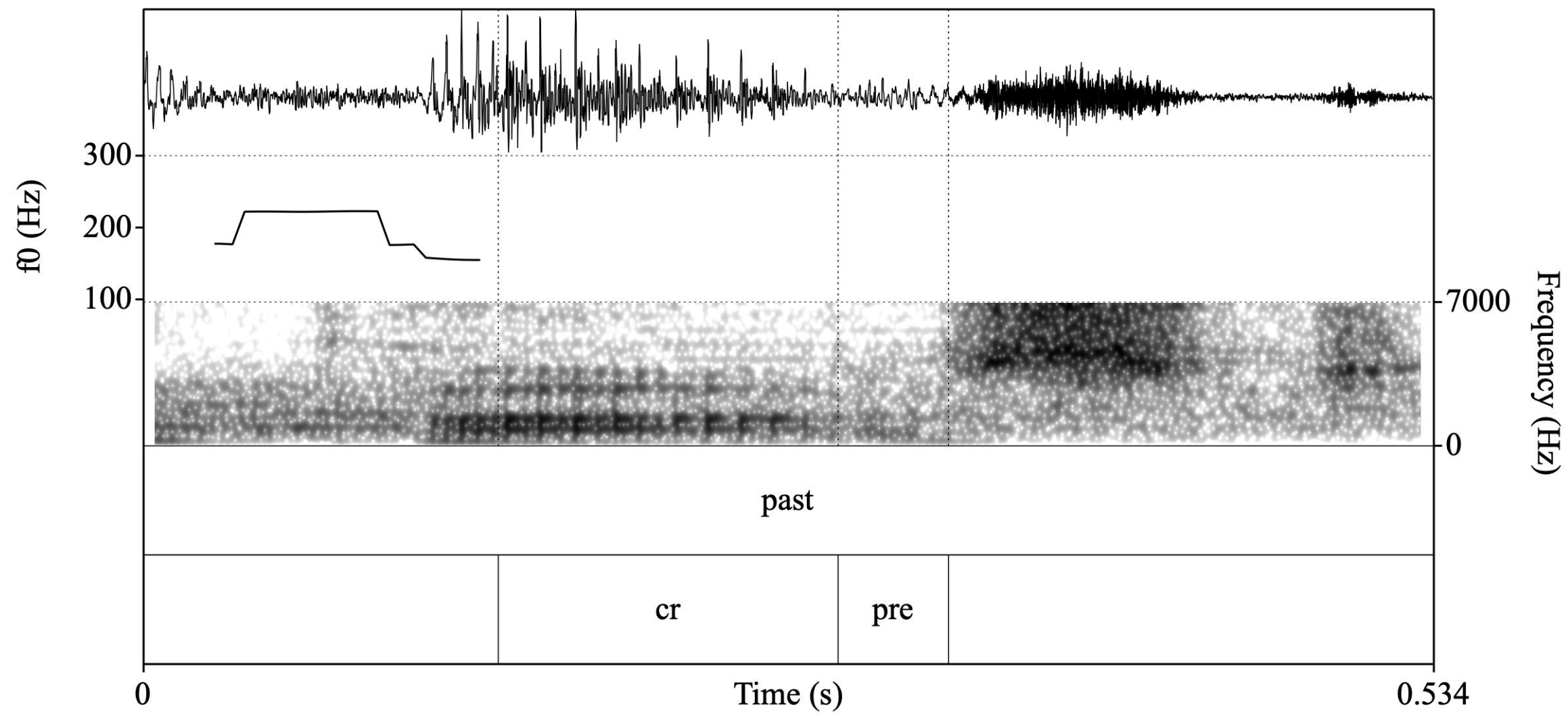
- Identified all possible pre-aspiration environments through 18 speeches from 1952 (age 26) to 2018 (age 92)
 - Stressed vowel + voiceless obstruent sequences: 2,061 tokens
- Voiceless **pre-aspiration** identified by presence of voiceless friction dispersed across different frequencies
- Local **breathiness** identified by an increase in glottal friction following the modal part of the vowel
- **Glottalization** identified as a period of irregular phonation or a sudden drop in f0
- Each phenomenon coded for binary presence/absence, since archival quality of data not high enough for exact durations



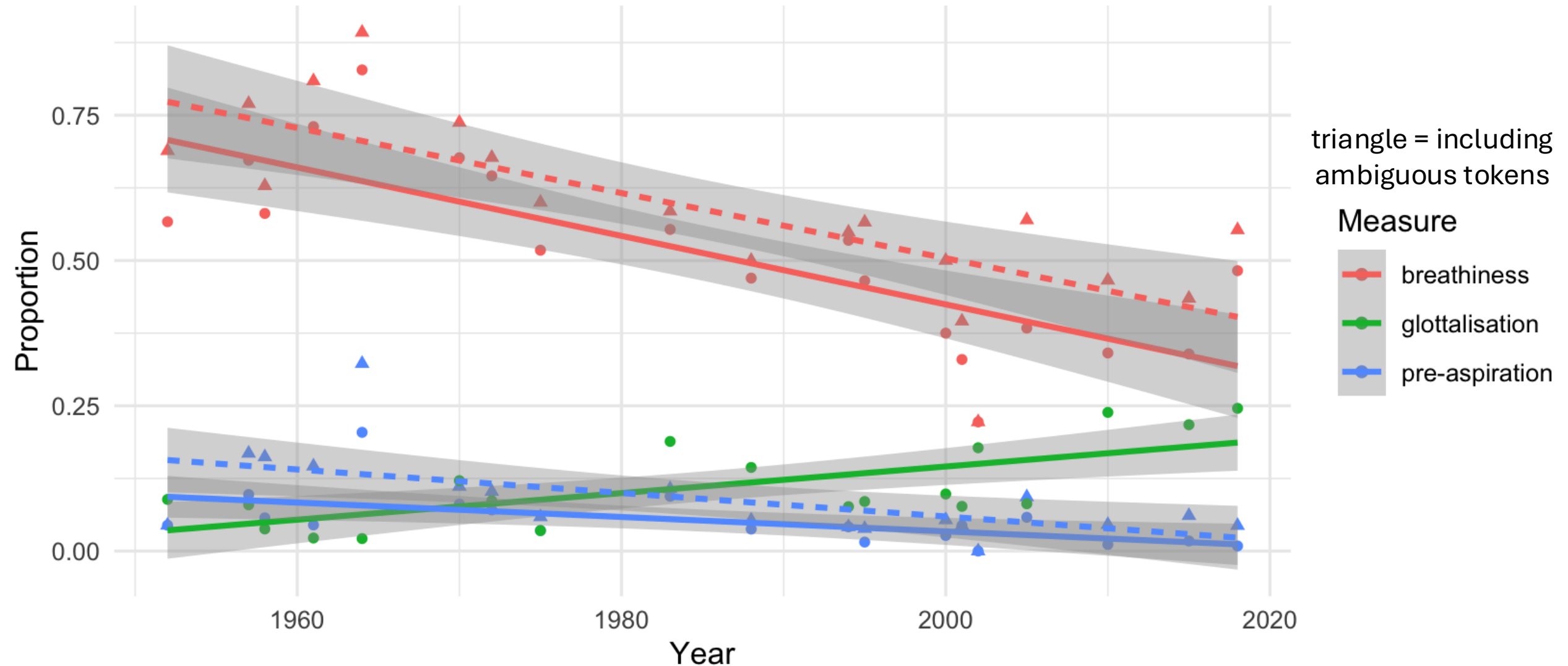




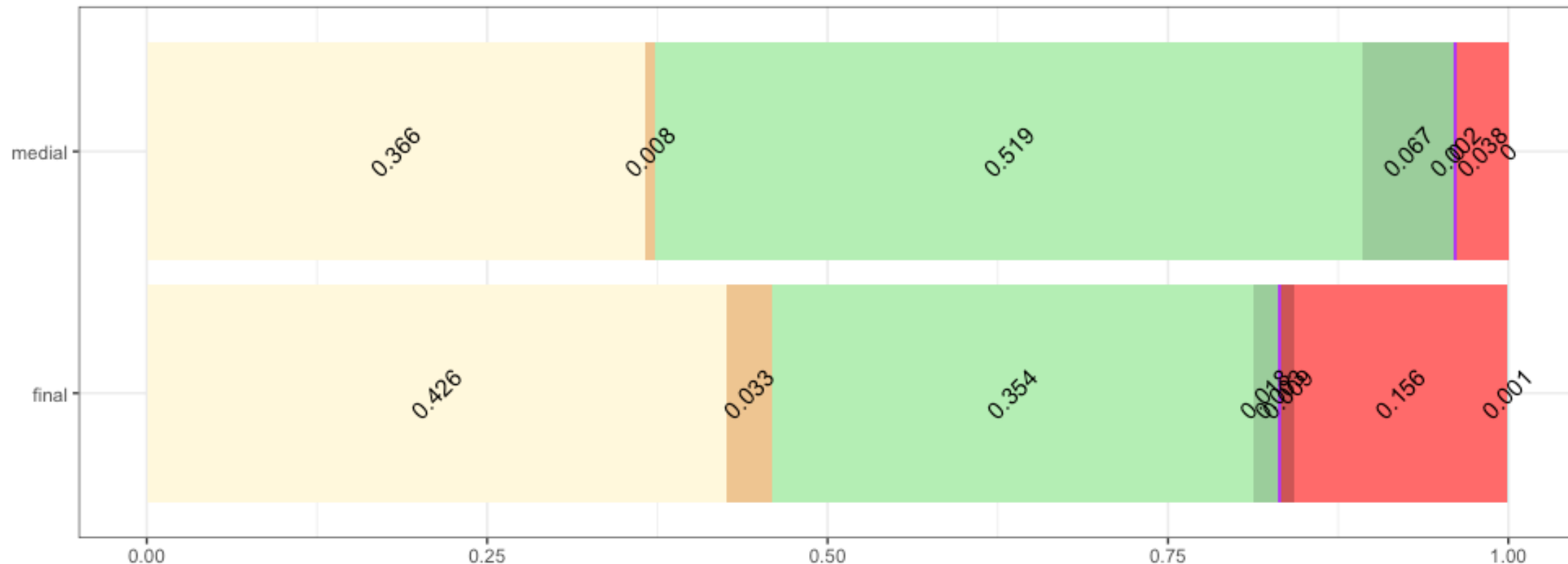




What the longitudinal data look like:



Co-occurrence of pre-aspiration, breathiness, and glottalisation within tokens

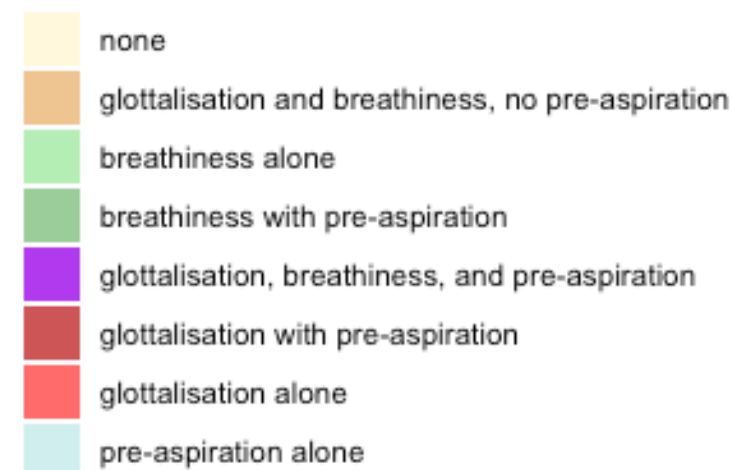


Foot-medial vs. foot-final makes a difference

Matty

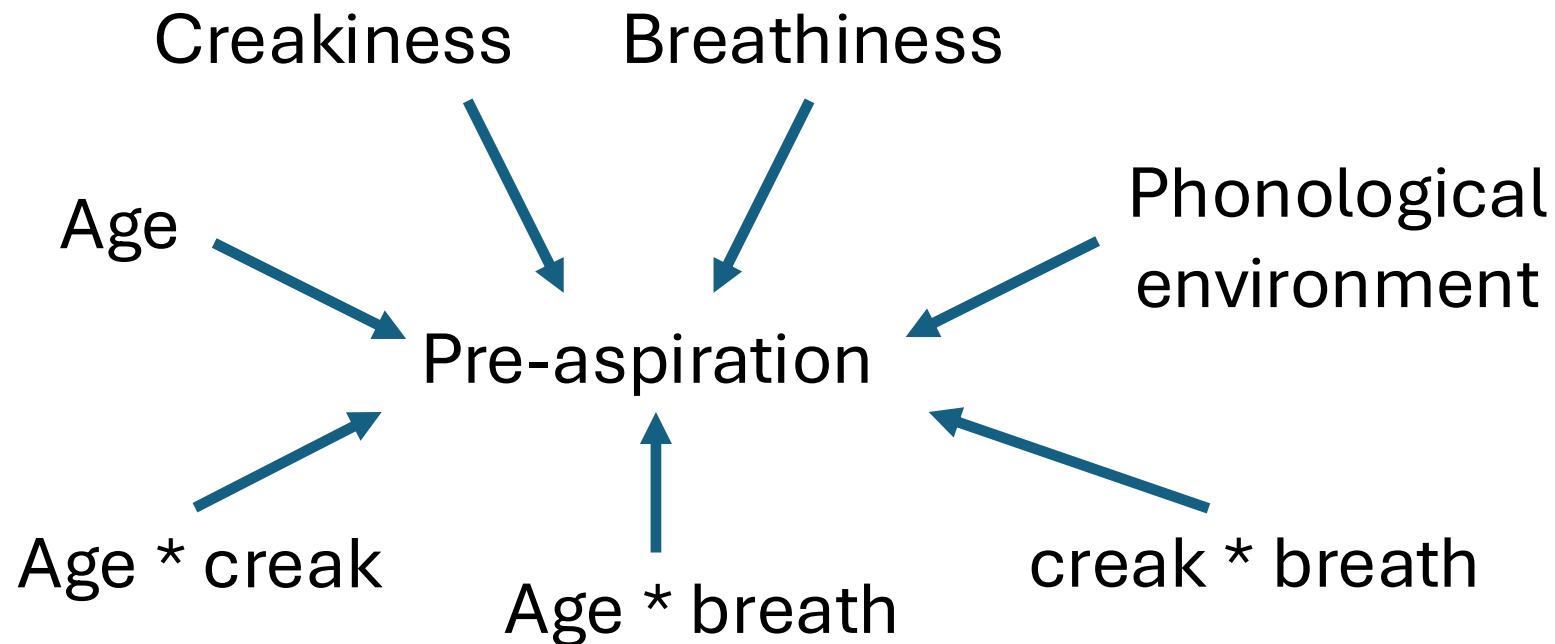
Matt

(Kettig & Hejná 2026)

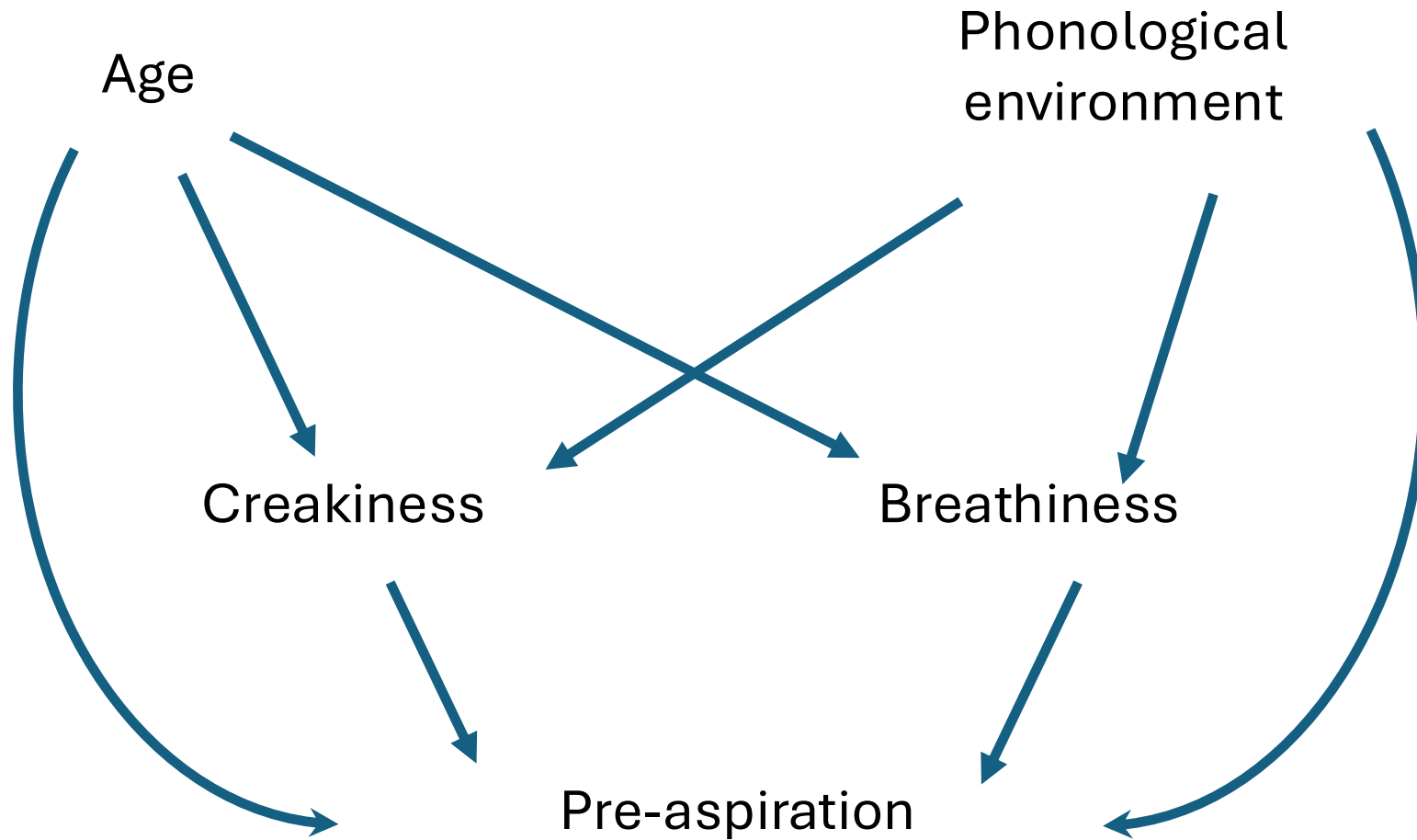


How we might be tempted to model it with glmer

`pre ~ age + creakiness + breathiness + environment + (1|word)`

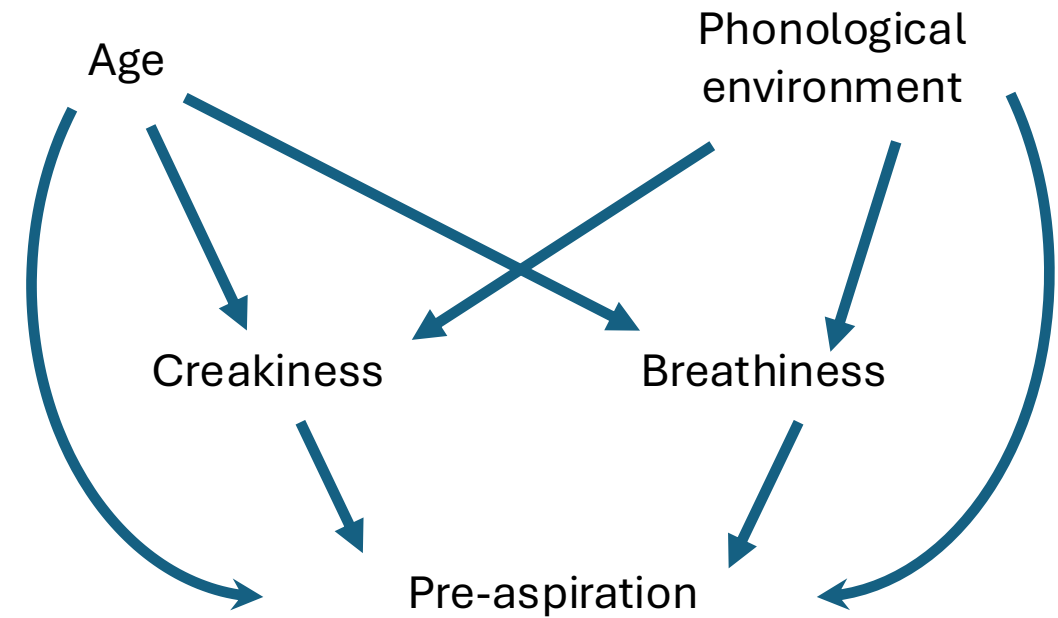


But what if we think there might be indirect effects?



But what if we think there might be indirect effects?

- Indirect effect: the portion of a predictor's (X) influence on an outcome (Y) that is explained by X influencing a mediator (M) which in turn influences Y (Agler & De Boeck 2017)
 - $X \rightarrow M$ = path a
 - $M \rightarrow Y$ = path b
 - Indirect effects are estimated as the product of coefficients of these paths (i.e., $a \times b$)
- Bayesian mediation (Yuan & McKinnon 2009) using brms package in R (Bürkner 2017)
- First estimate all direct effects in the model, then take posterior draws of those parameters to compute indirect effects and their credible intervals
- We use logit link function since all endogenous variables are binary here



Our SEM results for this dataset

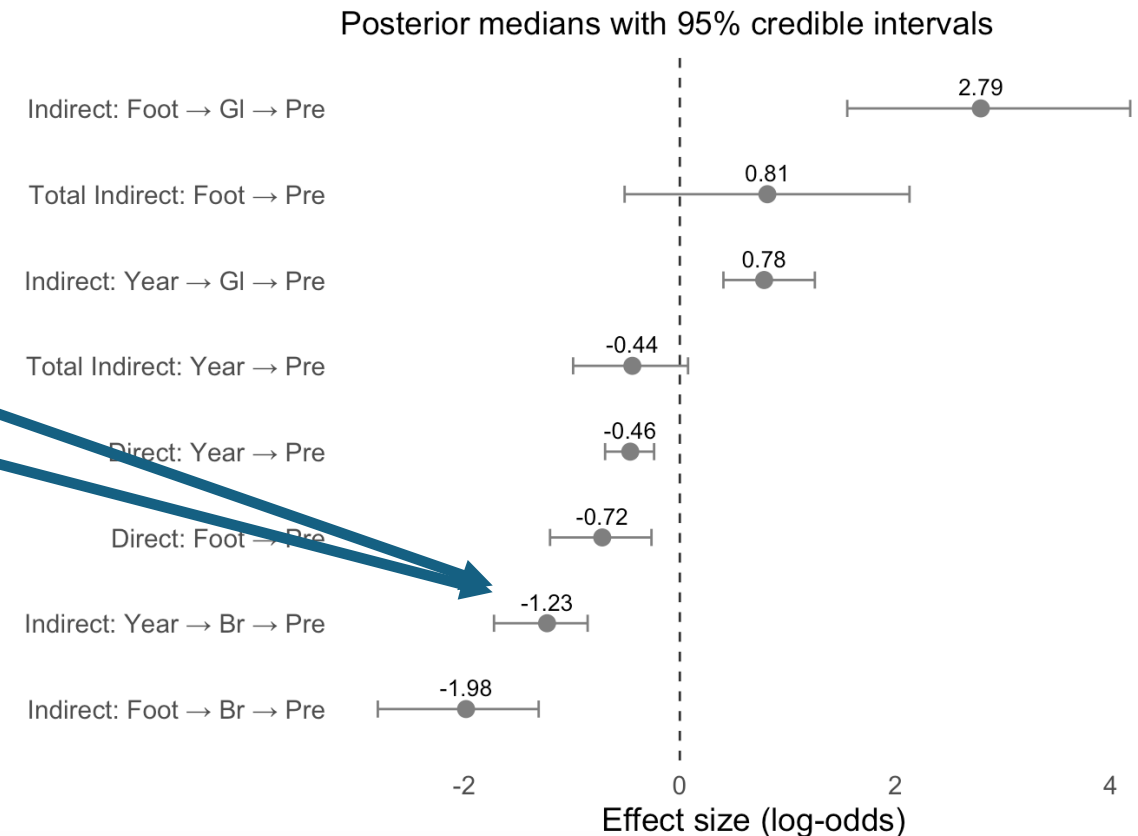
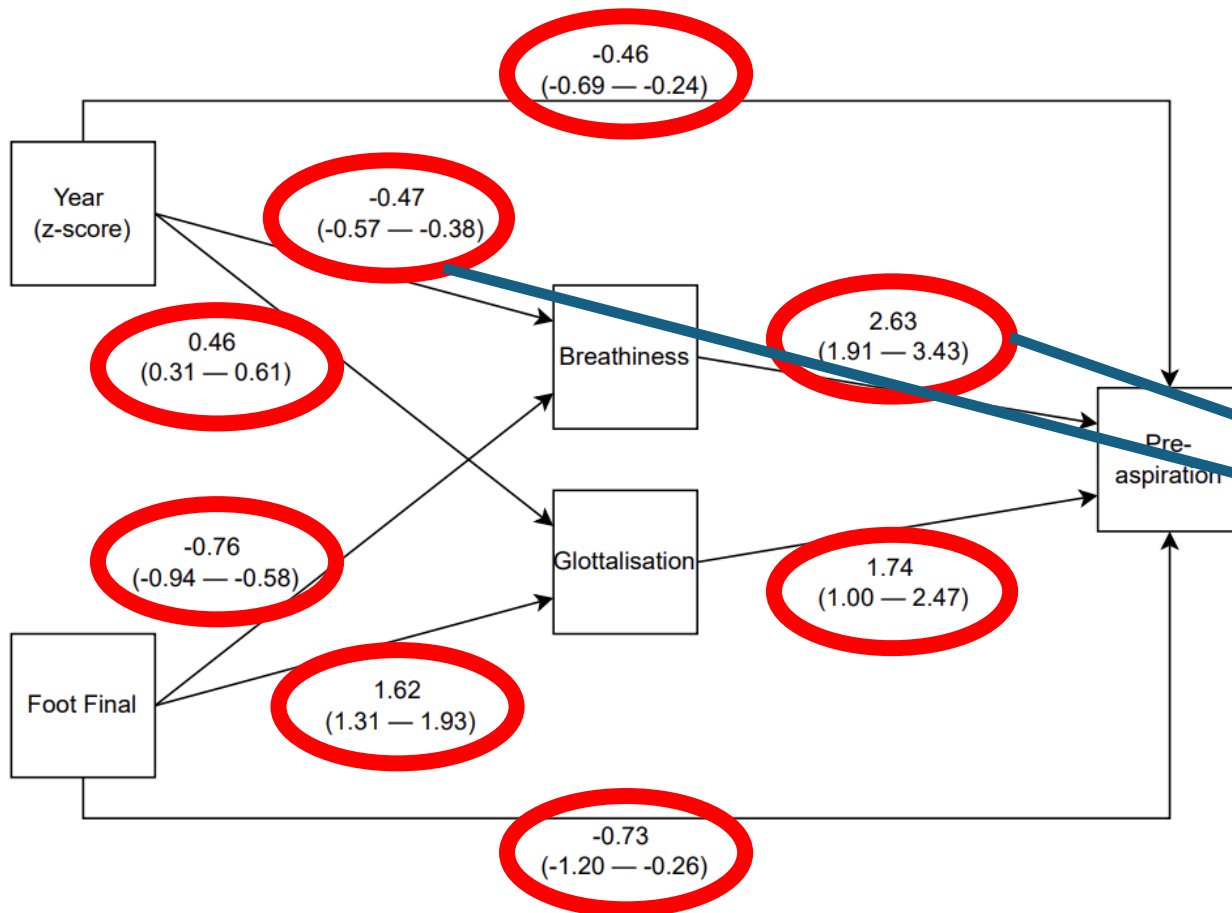
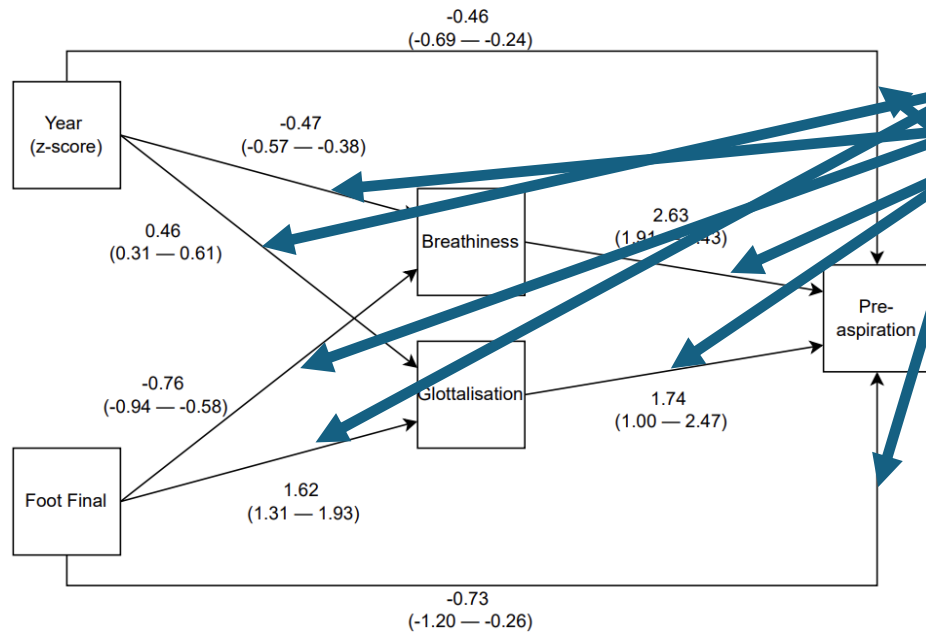
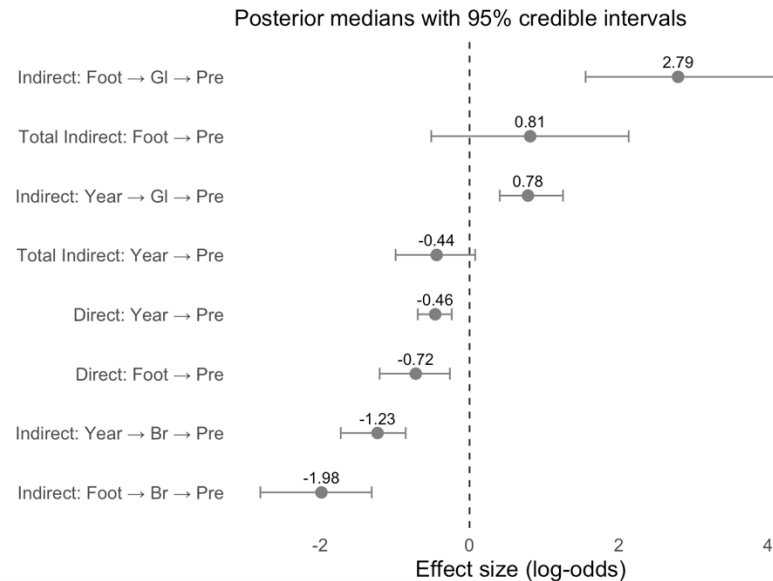


Fig. 1. Summary of direct effects within the model. Estimates represent log-odds with 95% credible intervals. **Fig. 2.** Summary of indirect effects within model. Estimates represent log-odds with 95% credible intervals.



```
# specific effects
bf_cr <- bf(cr ~ Year_scaled + foot_final, family = bernoulli(link = "logit"))
bf_br <- bf(br ~ Year_scaled + foot_final, family = bernoulli(link = "logit"))
bf_pre <- bf(pre ~ Year_scaled + foot_final + cr + br, family = bernoulli(link = "logit"))

# fit model
fit_brms_med_o <- brm(bf_cr + bf_br + bf_pre + set_rescor(FALSE),
  data = aggregated_data_NoInitial,
  chains = 4,
  cores = 4,
  iter = 4000,
  control = list(adapt_delta = 0.95))
```



posterior indirect effects (08/01/2026)

```
# extract posterior draws
posterior <- posterior_samples(fit_brms_med_o)

# compute indirect effects for each posterior draw
posterior <- posterior |>
  mutate(
    # indirect effects for Year
    indirect_cr_Year = b_pre_cr * b_cr_Year_scaled,
    indirect_br_Year = b_pre_br * b_br_Year_scaled,
    total_indirect_Year = indirect_cr_Year + indirect_br_Year,
    direct_Year = b_pre_Year_scaled,

    # indirect effects for foot_final
    indirect_cr_FF = b_pre_cr * b_cr_foot_final,
    indirect_br_FF = b_pre_br * b_br_foot_final,
    total_indirect_FF = indirect_cr_FF + indirect_br_FF,
    direct_FF = b_pre_foot_final)
```

Summary

- SEM and other mediated modelling approaches could help us better approach situations where we suspect indirect effects
 - Could especially be the case for lifespan changes, where aging might be more directly affecting one or more variables which in turn feed or bleed other phenomena
- Not many other people in phonetics doing this, but some have been using it, particularly for exploratory analyses
 - See work by Amanda Doucette, Jeanne Brown (McGill)
 - Let us know if you're aware of others doing mediated modeling!
- Our case study here has various limitations (binary coding, relatively small data, single speaker) but this approach should be able to handle lots of data structures and hypotheses

References

- Agler, R., & De Boeck, P. (2017). On the interpretation and use of mediation: Multiple perspectives on mediation analysis. *Frontiers in psychology*, 8, 1984.
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More details on SEM

- Structural Equation Modelling (SEM): set of methods to analyse systems of observed and/or latent variables (full-SEM includes latent and structural models; package lavaan in R)
 - Path analyses (a form of SEM without latent variables), implemented as a system of regression equations
 - Bayesian implementations also available (e.g., package brms)
- Assumptions
 - Correct causal ordering: $X \rightarrow M \rightarrow Y$
 - Requires theoretical justification
- Terms
 - Direct effects (DE): effect of X on Y controlling for M
 - Indirect effects (IE): effect transmitted via M
 - Total indirect effect (TIE): sum of indirect effects across multiple mediators (if multiple M)
 - Total effect (TE): overall effect of X on Y ($TE = DE + TIE$)

Audience questions:

- What do we do with our issues of independence, like the by-word and by-speaker random effects that we always put into our linear mixed-effects models?
 - In SEM, like other models, independence is usually assumed
 - Multilevel SEM (MSEM) is able to disentangle between and within level variances, and there are, e.g., design-based, corrections available that correct for dependencies within data structures

Audience questions:

- Unlike lmers, this doesn't have as many issues with collinearity of variables
 - Each effect is split across multiple paths instead of being forced into a single equation
 - Each coefficient is estimated for a specific relationship within the system of variables
 - As opposed to an interaction, this mediation shows how the effect is transmitted through a mediator

Consonant	Total medial	No. present (incl. ambig.)	% present (incl. ambig.)	Total final	No. present (incl. ambig.)	% present (incl. ambig.)
p	165	0 (5)	0% (3%)	67	0 (1)	0% (1.5%)
t	248	61 (74) <i>battle</i>	24.6% (29.8%)	265	7 (17)	2.6% (6.4%)
tʃ	44	2 (8)	4.5% (18.2%)	19	1 (1)	5.3% (5.3%)
k	148	4 (11)	2.7% (7.4%)	188	2 (5)	1.1% (2.7%)
f	81	4 (6)	4.9% (7.4%)	48	6 (12) <i>life</i>	12.5% (25%)
θ	14	1 (4) <i>nothing</i>	7.1% (28.6%)	69	4 (9)	5.8% (13%)
s	299	8 (14)	2.7% (4.7%)	233	8 (13)	3.4% (5.6%)
ʃ	160	0 (1)	0% (0.6%)	13	0 (0)	0% (0%)